

Yeshwantrao Chavan College of engineering, Nagpur.

Data Analytics Lab 04

Name :	Mrunali Jibhakate	Class/roll no :	6 th EE B - 107
Reg. No :	23071350		
Aim :	To perform data cleaning and implement categorical data encoding techniques, specifically Label Encoding and One-Hot Encoding, on a Titanic passenger dataset using the Python pandas and scikit-learn libraries.		
Implementation Steps :	• Step 1: Dataset Loading • Step 2: Handling Missing Values • Step 3: Label Encoding (Ordinal Data) • Step 4: One-Hot Encoding (Nominal Data) • Step 5: Data Concatenation		
Program Code :	<pre>import pandas as pd from sklearn.preprocessing import LabelEncoder # 1. Load dataset df = pd.read_csv('titanic_dataset.csv') # 2. Handle Missing Values df['Embarked'] = df['Embarked'].fillna(df['Embarked'].mode()[0]) # 3. Label Encoding for Pclass le = LabelEncoder() df['Pclass_Label'] = le.fit_transform(df['Pclass']) # 4. One-Hot Encoding for Embarked df_embarked_onehot = pd.get_dummies(df['Embarked'], prefix='Embarked') # 5. Combine and View Result df_final = pd.concat([df[['PassengerId', 'Pclass', 'Pclass_Label', 'Embarked']], df_embarked_onehot], axis=1) print("--- First 5 rows of encoded data ---") print(df_final.head()) 2) # Cleaning up for the model df_model = df_final.drop(['Embarked'], axis=1) df_model</pre>		

Yeshwantrao Chavan College of engineering, Nagpur.

Data Analytics Lab 04

Output :

```
--- First 5 rows of encoded data ---
  PassengerId  Pclass  Pclass_Label  Embarked  Embarked_C  Embarked_Q  \
0            1      2              1         Q         False      True
1            2      2              1         Q         False      True
2            3      1              0         C          True      False
3            4      1              0         S         False      False
4            5      2              1         Q         False      True

  Embarked_S
0         False
1         False
2         False
3          True
4         False
```

```
[10]:
```

	PassengerId	Pclass	Pclass_Label	Embarked_C	Embarked_Q	Embarked_S
0	1	2	1	False	True	False
1	2	2	1	False	True	False
2	3	1	0	True	False	False
3	4	1	0	False	False	True
4	5	2	1	False	True	False
...	...	...	...	...	...	...
2995	2996	3	2	False	False	True
2996	2997	1	0	False	True	False
2997	2998	3	2	False	False	True
2998	2999	3	2	True	False	False
2999	3000	1	0	True	False	False

3000 rows x 6 columns

Result :

Successfully implemented data preprocessing on the Titanic dataset. We concluded that **Label Encoding** is suitable for features with an inherent order, while **One-Hot Encoding** is essential for categorical features without a specific ranking to avoid bias in model training.

Marks :

Name of Lecturer : Dr. S.A. Dhondse

Signature :